Input system:
Dependent variables:$p_1, q_2, x_3, p_2, q_1, q_3, z_2, z_1, p_3, y_2, r_2, r_1, z_3, y_1, x_1, y_3, r_3, x_2$
Independent variables:t

$$\frac{dx_1}{dt} = p_1$$
$$\frac{dy_1}{dt} = r_1$$
$$\frac{dz_1}{dt} = q_1$$
$$\frac{dx_2}{dt} = p_2$$
$$\frac{dy_2}{dt} = r_2$$
$$\frac{dz_2}{dt} = q_2$$
$$\frac{dx_3}{dt} = p_3$$
$$\frac{dy_3}{dt} = r_3$$
$$\frac{dz_3}{dt} = q_3$$

$$\frac{dp_1}{dt} = Gm_2(x_2-x_1)\text{isqrt}((x_1-x_2)^2+(y_1-y_2)^2+(z_1-z_2)^2)^3+Gm_3(x_3-x_1)\text{isqrt}((x_1-x_3)^2+(y_1-y_3)^2+(z_1-z_3)^2$$
$$\frac{dr_1}{dt} = Gm_2(y_2-y_1)\text{isqrt}((x_1-x_2)^2+(y_1-y_2)^2+(z_1-z_2)^2)^3+Gm_3(y_3-y_1)\text{isqrt}((x_1-x_3)^2+(y_1-y_3)^2+(z_1-z_3)^2$$
$$\frac{dq_1}{dt} = Gm_2(z_2-z_1)\text{isqrt}((x_1-x_2)^2+(y_1-y_2)^2+(z_1-z_2)^2)^3+Gm_3(z_3-z_1)\text{isqrt}((x_1-x_3)^2+(y_1-y_3)^2+(z_1-z_3)^2$$
$$\frac{dp_2}{dt} = Gm_1(x_1-x_2)\text{isqrt}((x_1-x_2)^2+(y_1-y_2)^2+(z_1-z_2)^2)^3+Gm_3(x_3-x_2)\text{isqrt}((x_2-x_3)^2+(y_2-y_3)^2+(z_2-z_3)^2$$
$$\frac{dr_2}{dt} = Gm_1(y_1-y_2)\text{isqrt}((x_1-x_2)^2+(y_1-y_2)^2+(z_1-z_2)^2)^3+Gm_3(y_3-y_2)\text{isqrt}((x_2-x_3)^2+(y_2-y_3)^2+(z_2-z_3)^2$$
$$\frac{dq_2}{dt} = Gm_1(z_1-z_2)\text{isqrt}((x_1-x_2)^2+(y_1-y_2)^2+(z_1-z_2)^2)^3+Gm_3(z_3-z_2)\text{isqrt}((x_2-x_3)^2+(y_2-y_3)^2+(z_2-z_3)^2$$
$$\frac{dp_3}{dt} = Gm_1(x_1-x_3)\text{isqrt}((x_1-x_3)^2+(y_1-y_3)^2+(z_1-z_3)^2)^3+Gm_2(x_2-x_3)\text{isqrt}((x_2-x_3)^2+(y_2-y_3)^2+(z_2-z_3)^2$$
$$\frac{dr_3}{dt} = Gm_1(y_1-y_3)\text{isqrt}((x_1-x_3)^2+(y_1-y_3)^2+(z_1-z_3)^2)^3+Gm_2(y_2-y_3)\text{isqrt}((x_2-x_3)^2+(y_2-y_3)^2+(z_2-z_3)^2$$
$$\frac{dq_3}{dt} = Gm_1(z_1-z_3)\text{isqrt}((x_1-x_3)^2+(y_1-y_3)^2+(z_1-z_3)^2)^3+Gm_2(z_2-z_3)\text{isqrt}((x_2-x_3)^2+(y_2-y_3)^2+(z_2-z_3)^2$$

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Result:
Dependent variables:$x_3, p_2, q_1, z_1, r_2, r_1, y_1, r_3, p_1, q_3, s_3, z_2, z_3, p_3, y_2, s_1, q_2, s_2, x_1, y_3, x_2$
Independent variables:t

Additional variables:

$$s_1 = \text{isqrt}((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2)$$
$$s_2 = \text{isqrt}((x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2)$$
$$s_3 = \text{isqrt}((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2)$$

$$\frac{dx_1}{dt} = p_1$$
$$\frac{dy_1}{dt} = r_1$$
$$\frac{dz_1}{dt} = q_1$$
$$\frac{dx_2}{dt} = p_2$$
$$\frac{dy_2}{dt} = r_2$$
$$\frac{dz_2}{dt} = q_2$$
$$\frac{dx_3}{dt} = p_3$$
$$\frac{dy_3}{dt} = r_3$$
$$\frac{dz_3}{dt} = q_3$$

$$\frac{dp_1}{dt} = s_1^3 G m_2 x_2 + (-1) s_1^3 G m_2 x_1 + s_2^3 G m_3 x_3 + (-1) s_2^3 G m_3 x_1$$
$$\frac{dr_1}{dt} = s_1^3 G m_2 y_2 + (-1) s_1^3 G m_2 y_1 + s_2^3 G m_3 y_3 + (-1) s_2^3 G m_3 y_1$$
$$\frac{dq_1}{dt} = s_1^3 G m_2 z_2 + (-1) s_1^3 G m_2 z_1 + s_2^3 G m_3 z_3 + (-1) s_2^3 G m_3 z_1$$
$$\frac{dp_2}{dt} = s_1^3 G m_1 x_1 + (-1) s_1^3 G m_1 x_2 + s_3^3 G m_3 x_3 + (-1) s_3^3 G m_3 x_2$$
$$\frac{dr_2}{dt} = s_1^3 G m_1 y_1 + (-1) s_1^3 G m_1 y_2 + s_3^3 G m_3 y_3 + (-1) s_3^3 G m_3 y_2$$
$$\frac{dq_2}{dt} = s_1^3 G m_1 z_1 + (-1) s_1^3 G m_1 z_2 + s_3^3 G m_3 z_3 + (-1) s_3^3 G m_3 z_2$$
$$\frac{dp_3}{dt} = s_2^3 G m_1 x_1 + (-1) s_2^3 G m_1 x_3 + s_3^3 G m_2 x_2 + (-1) s_3^3 G m_2 x_3$$
$$\frac{dr_3}{dt} = s_2^3 G m_1 y_1 + (-1) s_2^3 G m_1 y_3 + s_3^3 G m_2 y_2 + (-1) s_3^3 G m_2 y_3$$
$$\frac{dq_3}{dt} = s_2^3 G m_1 z_1 + (-1) s_2^3 G m_1 z_3 + s_3^3 G m_2 z_2 + (-1) s_3^3 G m_2 z_3$$

$$\frac{ds_1}{dt} = (-1) s_1^3 p_1 x_1 + s_1^3 p_1 x_2 + s_1^3 p_2 x_1 + (-1) s_1^3 p_2 x_2 + (-1) s_1^3 r_1 y_1 + s_1^3 r_1 y_2 + s_1^3 r_2 y_1 + (-1) s_1^3 r_2 y_2 + (-1) s_1^3 q_1 z_1 + s_1^3 q_1$$
$$\frac{ds_2}{dt} = (-1) s_2^3 p_1 x_1 + s_2^3 p_1 x_3 + s_2^3 p_3 x_1 + (-1) s_2^3 p_3 x_3 + (-1) s_2^3 r_1 y_1 + s_2^3 r_1 y_3 + s_2^3 r_3 y_1 + (-1) s_2^3 r_3 y_3 + (-1) s_2^3 q_1 z_1 + s_2^3 q_1$$
$$\frac{ds_3}{dt} = (-1) s_3^3 p_2 x_2 + s_3^3 p_2 x_3 + s_3^3 p_3 x_2 + (-1) s_3^3 p_3 x_3 + (-1) s_3^3 r_2 y_2 + s_3^3 r_2 y_3 + s_3^3 r_3 y_2 + (-1) s_3^3 r_3 y_3 + (-1) s_3^3 q_2 z_2 + s_3^3 q_2$$